

Equal Weight per Instance Does Not Pay Alone: A Blob-Loss Auxiliary on Full-Resolution Cityscapes Buys Thin-Object IoU with Pedestrian Recall, and Only Pays as an Expert of a Mixture-of-Experts

Cityscapes val · ConvNeXt-V2-Base + UPerNet · CE + Dice + 0.5·blob (Kofler, IPMI 2023) vs CE + Dice · 3 seeds × 160 epochs at 1024×2048

Abstract

We report a pre-registered, controlled evaluation of the **blob loss** of Kofler *et al.* [IPMI 2023] — an auxiliary term that gives every ground-truth instance the same weight regardless of its pixel count — ported to **exclusive multi-class semantic segmentation at the native Cityscapes resolution** (1024×2048, 19 classes, softmax regime). Arm **G** (CE + Dice + 0.5·blob) is compared against its exact paired reference **B** (CE + Dice): same ConvNeXt-V2-Base + UPerNet architecture, same 160-epoch recipe, same augmentation distribution, three shared seeds (42, 123, 456) — only the loss changes.

The **pre-registered primary endpoint is null**: dataset-level official mIoU on the shared 500-image holdout, $\Delta(G-B) = +0.170$ pt, 95 % CI [-0.233 ; +0.551], two-sided paired image-bootstrap $p = 0.4032$ (B = 10 000 replicates). Equal weight per instance does **not** improve global mIoU in this regime. What the term actually does is a **measured trade-off**. It *buys* thin-object pixel coverage — traffic light +0.98, pole +0.58, bicycle +0.57 IoU vs B (Holm-significant within the 19-class family), truck +3.78 (CI excluding zero), Boundary F1 (3 px) +0.63 — and it *pays* in pedestrian instance integrity: strict pedestrian recall **-4.34**, small-instance stratum (T1) **-5.65**, crowd-group instances **-5.74**, pedestrian pixel precision **-2.70** (all vs paired B, Holm = 0), and a general **fragmentation** of the masks: connected components rise from 615.1 to 933.9 per image ($\times 1.52$), up in **16 of 19 classes** against the paired arm B (14 Holm-significant rises; the only material decrease is terrain, -1.8) and in **19 of 19** against the control arm (person masks $\times 2.2$) — a new per-class decomposition that *revises* the working mechanistic hypothesis of the program (the term does not prune small components; it creates holes). On the program’s pre-registered versatility criterion (36 endpoints $\times$ 13 arms), G is a **dominated specialist**: mean percentile 41.9, worst rank 13/13, maximal damage -6.76 pt, 13 significant losses against 4 significant gains.

The same verdict holds on a second dataset and a second probability regime: on BRATS 2023 (region-based sigmoid, MedNeXt, nnU-Net recipe, 5-fold CV, $n = 1\,196$), the blob arm alone scored **-0.00376 Dice** vs baseline, yet as **expert 3 of the winning MoE-V3** it contributed **+0.00566 Dice, first of 29 arms** [DOI 10.5281/zenodo.22903668]. The Cityscapes companion mixture **MoE-V3-CS** (four experts initialised from B, D, Dp, G; top-2 patch-wise gate) tells the same story: it absorbs G’s thin-object speciality — small-instance recall T1 +0.80 (significant), traffic light not degraded — without inheriting its damage (maximal damage -0.53 pt; Δ mIoU +0.45 pt, $p = 0.0066$ vs its paired control).

Contributions. (1) A pre-registered **null primary** for equal-weight-per-instance auxiliary loss in full-resolution exclusive-softmax segmentation, reported as such. (2) A complete **trade-off diagnostic**: per-class IoU, seven business-metric families on pedestrian instances, and a **new per-class connected-component decomposition** (933.9 vs 615.1 components/image) that contradicts the initially hypothesised compaction mechanism — the paper reports what the tables show. (3) An **exact port** of Kofler’s algebra (eq. 1) to the exclusive-softmax regime with precomputed instance packs: +1–2 % epoch time against +870 s/epoch for the naive path, naive $\leftrightarrow$ precomputed parity locked by unit tests, and the horizontal-flip alignment pitfall documented. (4) Evidence, on two datasets and two probability regimes, that the term **pays as an initialised expert of a mixture** while null-or-harmful alone. (5) Public release of code, configs, tables, figures and regeneration scripts.

1. Introduction

Voxel-wise losses — cross-entropy, Dice — weight every ground-truth instance by its volume. Small instances (rare classes, distant or partially occluded objects) contribute few pixels to the loss and are, in a strict optimisation sense, *rationaly sacrificed*. Kofler *et al.* [2023] proposed the **blob loss**: an auxiliary term that computes a soft binary dice **per ground-truth instance**, over a domain restricted to that instance plus all pixels outside other instances of the same class, and averages instances $\rightarrow$ class $\rightarrow$ image $\rightarrow$ batch. Every instance then carries the same total weight regardless of its pixel count. The term was validated in biomedical segmentation under the nnU-Net **region-based** recipe (overlapping WT/TC/ET targets, sigmoid channels), where it improved small-structure metrics.

This paper asks the transfer question on the other side of the fence:

Does equal weight per instance pay in **exclusive multi-class** semantic segmentation — softmax over 19 flat classes — at the **native Cityscapes resolution**, under a fixed training budget and a pre-registered primary endpoint?

The setting is a controlled arm of a four-paper loss program on Cityscapes at 1024×2048 with ConvNeXt-V2-Base + UPerNet: a distance-map auxiliary regression study [Cassez 2026a, DOI 10.5281/zenodo.21006236], a boundary-loss ablation whose arm B (CE + Dice, 160 epochs, seeds 42/123/456) is the exact paired reference used here [Cassez 2026b, DOI 10.5281/zenodo.21006393], this blob-loss study, and a companion mixture-of-experts report (MoE-V3-CS) in which the present arm G serves as expert 3. The program mirrors a completed BRATS track [DOI 10.5281/zenodo.22903668; 10.5281/zenodo.22906447; 10.5281/zenodo.22904810; 10.5281/zenodo.20110976], so every blob arm exists on both datasets under two probability regimes.

The answer, measured before this manuscript was written and pre-registered in the program’s artifact chain, is **no** — **the primary endpoint is null** ($\Delta \text{mIoU} +0.170 \text{ pt}$, $p = 0.4032$). This is therefore a negative-result paper in the house discipline of the program: negative and neutral results are reported as such, all official metrics are shown, and the contribution is the **diagnostic** (what the term buys, what it breaks, and the mechanism the tables actually support), the **exact port** (algebra, cost, pitfalls), and the **usage** (an initialised expert of a mixture, measured on two datasets).

2. Related work

Class imbalance in semantic segmentation. Cross-entropy with class re-weighting is the standard remedy; we use ISNS weights (inverse square-root of the number of samples) in *both* arms, so the comparison isolates the blob term on top of an already re-weighted CE. Dice loss [Milletari 2016] optimises region overlap and is class-balanced at the *class* level, not at the *instance* level: within a class, one large instance still dominates many small ones. Focal loss [Lin 2017] re-weights hard pixels, again pixel-wise. The blob loss is the imbalance remedy that operates **per instance**.

Blob loss. Kofler *et al.* [2023, IPMI; arXiv:2205.08209] introduce the term for biomedical segmentation under the nnU-Net region-based recipe [Isensee 2021], with total loss $\mathcal{L} = \mathcal{L}_{seg} + \beta \mathcal{L}_{blob}$ and the paper’s ratio $\alpha : \beta = 2 : 1$. Their validation regime is **sigmoid** (independent per-region channels, overlapping regions). The exclusive-softmax regime studied here puts classes in direct competition for every pixel; §7 discusses why the verdict differs from the intuition the term invites.

Boundary-aware and distance-based companions. Within the same program and hardware: the Kervadec boundary loss [Kervadec 2019] ablation [Cassez 2026b] and the distance-map auxiliary regression [Cassez 2026a] both report their pre-registered endpoints on the identical holdout and bootstrap protocol, which makes the present arm directly comparable to them (13-arm context table, §6.1).

Fragmentation and consensus filtering. Connected-component counts are a spatial-coherence proxy to which dataset-level mIoU is blind. The companion boundary paper measures a CC **consensus veto** ($C\oslash B$) that prunes −18.7 % of spurious fragments at no mIoU cost [Cassez 2026b]; on BRATS the analogous rule is published in [DOI 10.5281/zenodo.22904810]. The per-class fragment decomposition introduced here (§6.4) is, to our knowledge, the first measurement of how an instance-balancing loss moves fragmentation *per class* on Cityscapes.

Mixture-of-experts from initialised specialists. The BRATS MoE-V3 [DOI 10.5281/zenodo.22903668] trains a patch-wise gate over four experts initialised from independently trained specialist arms — including the blob arm. The Cityscapes companion MoE-V3-CS transposes the design (four experts B, D, Dp, G; top-2 gate over 3×3 patches; 80 epochs from expert weights). This paper cites both for the *usage* half of its thesis; the MoE-V3-CS results themselves are reported in the companion study, not re-measured here.

3. Method — the blob loss in the exclusive-softmax regime

Let Ω be the image domain, $C = 19$ the number of classes, $p_c(x) \in [0, 1]$ the **softmax** probability of class c at pixel x (so $\sum_c p_c(x) = 1$ — the regime difference with BRATS), and $y(x) \in \{0, \dots, 18, 255\}$ the ground-truth train id (255 = void/ignore). For class c , the ground-truth mask splits into connected **instances** $n \in I_c$ (8-connectivity on the label map; group annotations of the Cityscapes instance maps are handled at the business-metric level, §5.3, not inside the loss).

Per-instance domain. Following Kofler *et al.* eq. 1 exactly, the domain of instance n of class c is the whole image *minus the other instances of the same class*:

$$\Omega_n = (\{x : y(x) \neq 255\} \setminus \bigcup_{m \in I_c, m \neq n} m) \cup n.$$

Per-instance soft binary dice. With t_n the indicator of instance n , $s_n = \sum_{x \in n} p_c(x)$ the predicted mass on the instance, $c_n = |n|$ its pixel count, and $S_0^{(c)} = \sum_{x \in \Omega_n \setminus n} p_c(x)$ the mass on the instance-free part of its domain (so $\sum_{x \in \Omega_n} p_c = S_0^{(c)} + s_n$):

$$\ell_n = 1 - \frac{2s_n + \varepsilon}{S_0^{(c)} + s_n + c_n + \varepsilon}, \quad \varepsilon = 1.0 \text{ (nnU-Net convention).}$$

Ignored pixels (255) fall in a trash bucket: they are excluded from both $S_0^{(c)}$ and the instances, and are never counted as false negatives (the valid semantics of the BRATS port).

Two-level mean. Faithful to the BRATS RegionBlobLoss the port comes from:

$$\mathcal{L}_{blob} = \frac{1}{|\mathcal{B}|} \sum_{img \in \mathcal{B}} \frac{1}{|C_{img}|} \sum_{c \in C_{img}} \frac{1}{|I_c|} \sum_{n \in I_c} \ell_n,$$

where C_{img} is the set of classes present in the image. Every instance of every present class carries the same total weight per image — the defining property of the term. `min_blob_pixels = 0`: no instance is filtered by size, since small instances are precisely what the term is meant to rescue.

Composite losses (arms G and B). With CE at ISNS class weights and Dice (smooth 1.0), both ignoring 255:

$$\mathcal{L}_G = 0.5 \mathcal{L}_{CE}^{ISNS} + 0.5 \mathcal{L}_{Dice} + 0.5 \mathcal{L}_{blob}, \quad \mathcal{L}_B = 0.5 \mathcal{L}_{CE}^{ISNS} + 0.5 \mathcal{L}_{Dice}.$$

$\beta = 0.5$ is Kofler’s default ($\alpha : \beta = 2 : 1$ with the segmentation loss weighing 1.0 in total). **No sweep of β is performed** — the fixed-budget comparison stance of the whole program (same position as the companion papers on λ_b and the distance-map weight).

4. Implementation — precomputed instance packs

Two execution paths, same algebra. (1) *Precomputed* (used for training arm G): per-image instance labels are packed offline into `.blob.npz` files (`blob_glob/blob_counts/blob_csr`, built by `src/losses/blob_lab.py` via `scripts/precompute_blob_lab.py`) and delivered by the dataloader; the forward pass is then elementary GPU scatter-add kernels — the CPU connected-component pass and the 26 small per-class GPU synchronisations disappear from the training loop. (2) *On-the-fly fallback*: without a pack, the same packing functions recompute it from the target — this is the correctness reference used by tests and evaluation harnesses, giving structural parity between the two paths.

Measured cost (RTX PRO 6000 Blackwell, full-resolution ground truth, 2026-09-17): the naive on-the-fly path costs **+870 s/epoch** — unacceptable; the precomputed path costs **+1–2 %/epoch**. The connected-component labelling itself runs at **22.4 ms/image** with `cc3d` (8-connectivity; 16 classes present, 118 instances on a representative full-resolution image) against 59.7 ms with `scipy.ndimage.label`. Packs are computed once for the 2 975-image train split (labels are shared across seeds).

Parity and regeneration. Naive↔precomputed parity is locked by `tests/test_blob_loss.py`. All 2 975 packs were regenerated **bit-identical** on 2026-09-26 (verified over the full set). The packs themselves are heavy and are not shipped in the release bundle; `scripts/precompute_blob_lab.py` regenerates them identically.

The flip pitfall (documented). The instance pack must undergo the **same** horizontal flip as image and label. In arm G the dataset applies the flip itself ($p = 0.5$, jointly on image + label + pack) and the augmentations flip is **disabled** in the config, so the blob glob can never be misaligned. The augmentation *distribution* is identical to arm B’s (hflip $p = 0.5$, photometric jitter, Gaussian blur); only the application point differs. A sanity inherited from the program’s SDT audit: all three resolved G configs have the structural flip OFF — the arm is unaffected by the misalignment bug found in earlier boundary/distmap arms.

5. Experimental setup

5.1 Model, recipe, budget

Architecture (identical in both arms): ConvNeXt-V2-Base [Woo 2023] pretrained ImageNet-22K (FCMAE) then fine-tuned on ImageNet-1K, + UPerNet head [Xiao 2018] with auxiliary FCN deep supervision (weight 0.4), 19 logits per pixel at full input

resolution. **Training:** 160 epochs, batch $2 \times$ gradient accumulation 4 (effective 8), AdamW (lr 6×10^{-5} , weight decay 0.01), polynomial decay (power 1.0, 1-epoch warmup), BF16 autocast, channels_last, inputs at native 1024×2048 without cropping. Seeds **42, 123, 456** in both arms. Arm G trained 17–20 September 2026.

Measured cost of arm G (per seed, `results/p3_queue/pilot_fullres_G_blob_seed{42,123,456}.log`): 160/160 epochs, median **656 s/epoch** [655–669], VRAM **31.5 GB**, final train loss 0.3743 / 0.3735 / 0.3765, $\approx$ **29.2 h per seed**, $\approx$ **87 h GPU** for the arm.

5.2 Data and holdout

Cityscapes fine annotations [Cordts 2016]: 2 975 train / 500 val images, 19 evaluation classes. All evaluations run on the program’s **pre-specified shared holdout**: the first 500 val images in loader order (`first:500`) — identical for all 13 arms of the program, fixed before any arm was trained. The official Cityscapes leaderboard was **not** submitted to (declared limitation, §8).

5.3 Metrics

- **mIoU** — dataset-level, from a single aggregated confusion matrix, estimator validated **bit-identical to the official cityscapesScripts** routine (`tests/test_official_miou.py`). Per-class IoU from the same matrices.
- **Boundary F1 (3 px)** — per-class F1 of predicted vs ground-truth contours within a 3-pixel tolerance [protocol of Perazzi 2016], averaged over classes present per image; also restricted to {person, rider}.
- **Pedestrian instance metrics** — from the official `*_gtFine_instanceIds.png` maps, classes {person, rider}: pixel recall *any-ped* (prediction $\in$ {person, rider}) and *strict* (prediction = the instance’s own class); detection rate at recall ≥ 0.5 ; pedestrian **pixel precision** (fraction of predicted-pedestrian pixels falling inside a ground-truth pedestrian instance).
- **Strata** — all instances / individual instances (`instId  $\geq$  1000`) / **crowd-group** instances (`instId < 1000` — the official ambiguous/tightly-packed pedestrian groups, the best available proxy for partially occluded pedestrians) / **size terciles** of individual instances, T1 = smallest tercile (proxy for distance and occlusion), T3 = largest.
- **Fragments** — 8-connected components per image summed over the 19 class masks (`count_fragments, src/postprocessing/consolidate.py`) plus a **new per-class decomposition** introduced by this paper (components per class per image, arms G/B/control).

5.4 Pre-registered primary endpoint and statistics

The single pre-registered primary endpoint is the **dataset-level official mIoU, arm G vs its paired reference B** (paired = same architecture, recipe, budget, augmentation, seeds — only the loss differs). Test: **paired image-bootstrap** over the 500 holdout images — positions resampled with replacement, **B = 10 000** replicates, bootstrap seed **20260618**, the three seeds averaged *within each replicate*, 95 % percentile CI, two-sided $p = 2 \cdot \min(\text{frac } \Delta \leq 0, \text{frac } \Delta \geq 0)$. The primary family is the single pair, so Holm = p .

Secondary/exploratory families, each with its Holm correction declared where reported: per-class IoU (19-class family per arm pair), business metrics (Holm(2) over this paper’s two pairs G-vs-B and G-vs-control), per-class fragmentation (19-class family), 13-arm mIoU context (the program’s 15-pair family, computed over the master table `results/moe_v3_cs/p314/master_table.json`).

Two significance criteria, never conflated. *Holm-significant* (corrected $p < 0.05$ within the stated family) and *CI-excluding-zero* (raw 95 % CI) are reported **separately** everywhere — bold in tables marks Holm significance only, and the p / Holm columns carry the exact values. This distinction matters for this arm: e.g. truck +3.78 IoU vs B excludes zero but does not survive the 19-class Holm (0.460), while traffic light +0.98 survives (Holm < 0.001).

The two comparison pairs. The budget-paired reference is **B** (160 epochs) — the pair of the primary endpoint. The 13-arm program context uses a different reference, the MoE **control** (CE + Dice, 80 epochs, B-initialised) — *not* budget-paired with G, always labelled as context, never substituted for the paired pair. Both are reported side by side throughout.

Non-determinism, declared. cuDNN benchmark is on (`deterministic: false`, BF16). The primary numbers come from the harness forward of the pre-registered artifact; the business-metric regeneration (§6.3) comes from a second independent GPU forward of the *same* checkpoints. The two agree within $\leq$ **0.01 pt** on every shared quantity (per-seed mIoU gaps $\leq 2 \times 10^{-4}$ pt; primary $\Delta +0.170$ vs $+0.173$, p 0.4032 vs 0.3952 — same verdict), of the same order as the 4.1×10^{-3} pt already documented in the program’s consolidation sanity.

6. Results

6.1 Primary endpoint: null

Pre-registered primary, paired image-bootstrap over the 500-image holdout (artifact results/moe_v3_cs/harness/table_Gseul_P310.js 2026-09-22):

Quantity	G — CE+Dice+0.5·blob	B — CE+Dice
mIoU (point)	81.263	81.093
95 % CI	[79.815 ; 82.431]	[79.694 ; 82.204]
mIoU seed 42	81.297	81.429
mIoU seed 123	81.291	80.861
mIoU seed 456	81.202	80.989

$\Delta(\text{G-B}) = +0.170 \text{ pt} \cdot 95 \% \text{ CI } [-0.233 ; +0.551] \cdot p = 0.4032$ (single pair, Holm = p) → **not significant**.

The honest reading: the primary endpoint is **null**. The blob loss does not improve mIoU in full-resolution exclusive-softmax semantic segmentation. Per-seed values do not hide a consistent direction either: G wins seed 123 (+0.43) and seed 456 (+0.21), loses seed 42 (-0.13).

The second, independent GPU forward of the same checkpoints (business-metric regeneration path, §6.3) gives $\Delta = +0.173 \text{ pt}$ $[-0.229 ; +0.554]$, $p = 0.3952$ — same verdict, $\text{gap} \leq 0.01 \text{ pt}$ (declared cuDNN/BF16 non-determinism).

13-arm program context (reference: MoE control, CE + Dice 80 epochs — *not* budget-paired with G; artifact results/moe_v3_cs/p314/main)

#	Arm	mIoU	Δ vs control	p	Holm (15 pairs)
1	D · CE+Kervadec EDT	81.69	+0.52 [+0.14 ; +0.91]	0.0048	0.058
2	consensus D⊗B	81.65	+0.48 [+0.11 ; +0.87]	0.0082	0.082
3	consensus Dp⊗B	81.65	+0.48 [+0.02 ; +0.96]	0.0396	0.317
4	Dp · CE+SDT (distmap)	81.64	+0.47 [+0.04 ; +0.93]	0.0312	0.281
5	MoE-V3-CS (4 experts)	81.62	+0.45 [+0.11 ; +0.82]	0.0066	0.073
6	A · CE alone	81.28	+0.11 [-0.27 ; +0.52]	0.5722	1.000
7	G · CE+Dice+Blob (this paper)	81.26	+0.10 [-0.28 ; +0.47]	0.6552	1.000
8	consensus C⊗B	81.24	+0.07 [-0.34 ; +0.47]	0.7758	1.000
9	C · CE+Dice+Kervadec EDT	81.23	+0.06 [-0.34 ; +0.47]	0.8096	1.000
10	B · CE+Dice (160 ep)	81.09	-0.08 [-0.45 ; +0.28]	0.6748	1.000
11	consensus Cp⊗B	81.01	-0.15 [-0.52 ; +0.17]	0.3498	1.000
12	Cp · CE+Dice+SDT	80.89	-0.28 [-0.60 ; +0.02]	0.0628	0.440
—	control (ref, CE+Dice 80 ep)	81.17	—	—	—

G ranks **7th of 13** by amplitude. No arm passes Holm 0.05 on the 15-pair exploratory family — stated as-is: on mIoU, the conclusions of this program rest on **amplitudes and their CIs**, not on post-Holm significance.

6.2 Per-class IoU: thin-object coverage bought

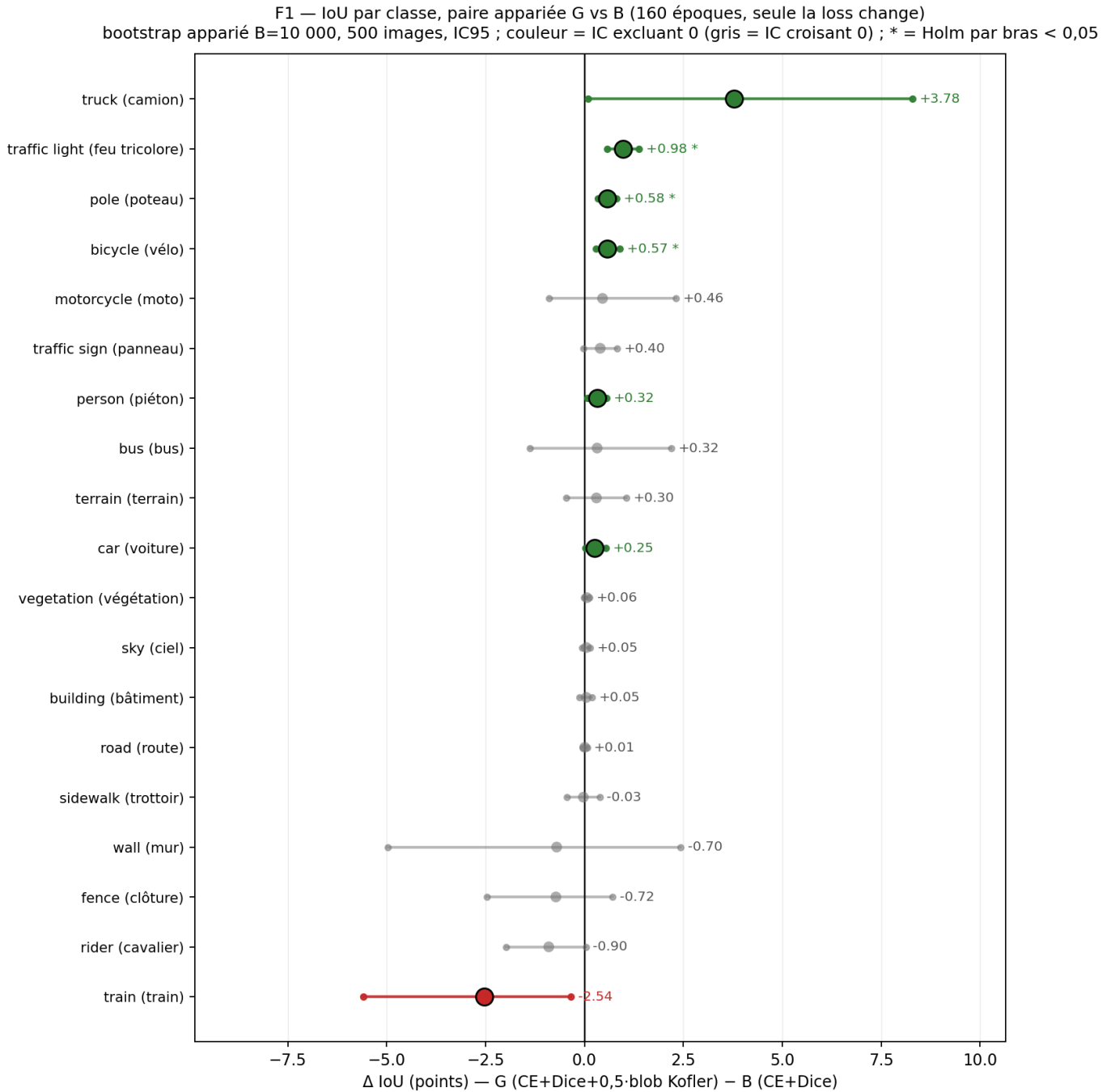


Figure 1: F1 — Per-class forest plot, G vs B paired

Figure 1: per-class Δ IoU, arm G vs paired reference B (bootstrap apparié, B = 10 000). Filled markers = Holm-significant within the 19-class family; open markers = CI excluding zero without Holm survival; the two criteria are plotted separately, never merged.

Full 19-class table (paired G-B; context G-control from the same artifact family). Bold = Holm (19-class family) < 0.05:

Class	IoU G	IoU B	$\Delta(G-B)$	95 % CI	p	Holm	$\Delta(G-ctrl)$	95 % CI	p	Holm
road	98.45	98.44	+0.01	[-0.05 ; +0.06]	0.7970	1.000	+0.04	[-0.02 ; +0.10]	0.1874	1.000
sidewalk	87.04	87.07	-0.03	[-0.44 ; +0.38]	0.9184	1.000	+0.06	[-0.31 ; +0.48]	0.7642	1.000
building	93.32	93.27	+0.05	[-0.13 ; +0.18]	0.5170	1.000	+0.07	[-0.03 ; +0.18]	0.1612	1.000
wall	57.21	57.92	-0.70	[-4.97 ; +2.43]	0.7974	1.000	+1.07	[-1.32 ; +4.05]	0.4916	1.000
fence	65.35	66.07	-0.72	[-2.47 ; +0.71]	0.3874	1.000	-0.46	[-1.29 ; +0.34]	0.2668	1.000
pole	70.34	69.76	+0.58	[+0.34 ; +0.81]	<0.0001	<0.001	+0.29	[+0.06 ; +0.51]	0.0136	0.231
traffic light	77.83	76.85	+0.98	[+0.58 ; +1.37]	<0.0001	<0.001	+0.47	[+0.12 ; +0.82]	0.0108	0.194
traffic sign	83.92	83.52	+0.40	[-0.04 ; +0.83]	0.0686	0.706	+0.21	[-0.13 ; +0.54]	0.2182	1.000
vegetation	93.15	93.09	+0.06	[-0.00 ; +0.13]	0.0588	0.706	+0.02	[-0.03 ; +0.07]	0.4260	1.000
terrain	66.40	66.10	+0.30	[-0.46 ; +1.06]	0.4454	1.000	+0.01	[-0.61 ; +0.60]	0.9850	1.000
sky	95.64	95.60	+0.05	[-0.05 ; +0.15]	0.3538	1.000	+0.11	[+0.00 ; +0.22]	0.0410	0.656
person	85.22	84.90	+0.32	[+0.07 ; +0.56]	0.0146	0.234	+0.07	[-0.18 ; +0.30]	0.5824	1.000
rider	67.58	68.49	-0.90	[-1.98 ; +0.03]	0.0632	0.706	-1.34	[-2.42 ; -0.35]	0.0072	0.137
car	95.71	95.46	+0.25	[+0.03 ; +0.54]	0.0212	0.297	+0.14	[-0.13 ; +0.45]	0.3136	1.000
truck	84.03	80.24	+3.78	[+0.09 ; +8.30]	0.0354	0.460	+3.43	[-1.22 ; +8.31]	0.1826	1.000
bus	90.07	89.75	+0.32	[-1.38 ; +2.20]	0.7154	1.000	-1.48	[-4.74 ; +1.01]	0.3274	1.000
train	81.20	83.74	-2.54	[-5.59 ; -0.34]	0.0176	0.264	-1.04	[-2.79 ; +0.86]	0.2966	1.000
motorcycle	71.15	70.70	+0.46	[-0.90 ; +2.31]	0.5112	1.000	+0.07	[-0.93 ; +0.99]	0.9362	1.000
bicycle	80.38	79.81	+0.57	[+0.28 ; +0.90]	<0.0001	<0.001	+0.05	[-0.17 ; +0.29]	0.6634	1.000

Against paired **B**: the Holm survivors are exactly the thin, signal-rich classes — **traffic light +0.98, pole +0.58, bicycle +0.57**. Truck (+3.78), person (+0.32) and car (+0.25) exclude zero but do not survive the 19-class Holm (0.460 / 0.234 / 0.297); train (-2.54) is the single degradation excluding zero (Holm 0.264). Against the **control**, G is the **only arm of the 13** that improves traffic light (+0.47, CI excluding zero; the boundary/distmap arms all *degrade* it by -2.3 to -2.8), at the price of rider (-1.34, CI excluding zero; Holm(2) 0.0136 in §6.3).

6.3 Business metrics: pedestrian integrity paid

F2 — Le trade-off du blob loss seul : IoU pixel des objets fins † contre rappel instance piéton ‡ — et la facture topologie : plus de composantes connexes dans 16 classes sur 19 vs B apparié (19 sur 19 vs contrôle, piétons ×2,2) G vs B apparié (160 époques, seule la loss change) ; gauche : attribution per-class P3.16 · milieu : métriques instance régénérées · droite : décomposition par classe des composantes connexes (mesure inédite, preds P3.16)

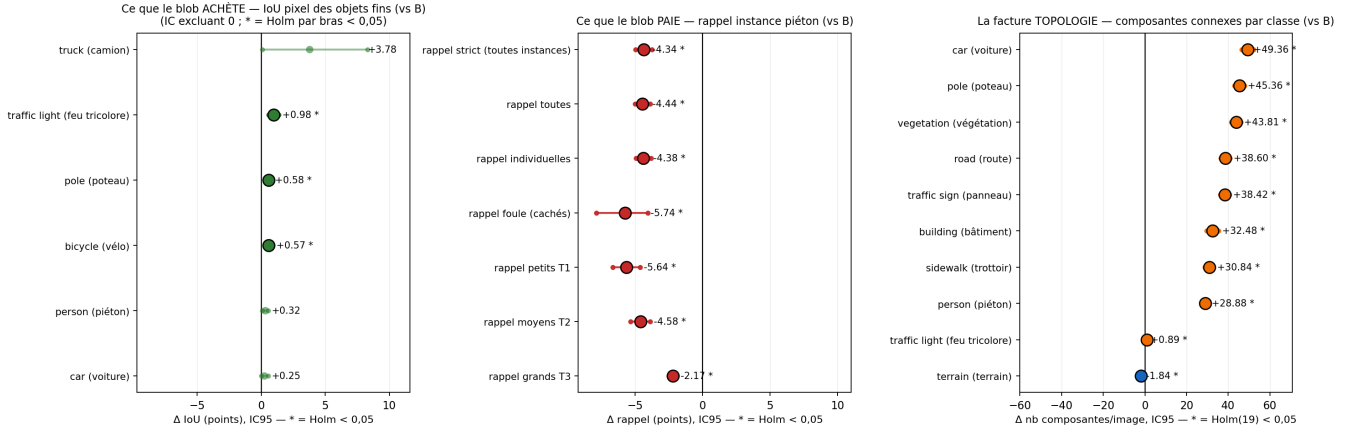


Figure 2: F2 — The measured trade-off

Figure 2: the trade-off, both pairs. Left/middle: thin-object pixel coverage (Boundary F1, per-class IoU) goes up; right: pedestrian instance recall (strict, size strata, crowd) and pedestrian pixel precision go down. CIs from the paired image-bootstrap.

Regenerated from the per-image npz artifacts (500 images × 3 seeds per arm; bit-identical to the program's cross-arm attribution artifacts on 19 shared metrics; ≤ 0.01 pt vs the primary harness forward on the shared mIoU row — 14/14 blocking sanity checks, papers/paper4/tables/SANITY.md). Units: points (×100), except fragments = components per image. **Bold = Holm(2) < 0.05** over this paper's two pairs; n = images contributing to the metric:

Metric	n	G	B	ctrl	Δ(G-B) [95 % CI]	p	Holm(2)	Δ(G-ctrl) [95 % CI]	p	Holm(2)
mIoU	500	81.26	81.09	81.17	+0.173 [-0.229 ; +0.554]	0.3952	0.7904	+0.094 [-0.281 ; +0.466]	0.6592	0.7904
IoU person	500	85.22	84.88	85.15	+0.344 [+0.092 ; +0.578]	0.0088	0.0176	+0.067 [-0.182 ; +0.302]	0.5838	0.5838
IoU rider	500	67.58	68.49	68.92	-0.911 [-2.003 ; +0.036]	0.0612	0.0612	-1.339 [-2.413 ; -0.352]	0.0068	0.0136
Boundary F1 3 px	500	77.00	76.37	76.63	+0.630 [+0.454 ; +0.806]	<0.0001	<0.001	+0.376 [+0.216 ; +0.537]	<0.0001	<0.001
Boundary F1 3 px, ped	500	67.73	67.30	67.49	+0.431 [-0.119 ; +0.953]	0.1204	0.2408	+0.247 [-0.201 ; +0.702]	0.2814	0.2814
Ped. strict recall	441	72.84	77.17	76.25	-4.335 [-4.948 ; -3.729]	<0.0001	<0.001	-3.417 [-3.967 ; -2.868]	<0.0001	<0.001
Ped. pixel precision	473	63.27	65.96	70.03	-2.699 [-3.484 ; -1.951]	<0.0001	<0.001	-6.763 [-7.800 ; -5.785]	<0.0001	<0.001
All- instances recall	441	76.55	80.99	79.98	-4.439 [-5.000 ; -3.884]	<0.0001	<0.001	-3.423 [-3.911 ; -2.942]	<0.0001	<0.001

Metric	n	G	B	ctrl	$\Delta(G-B)$ [95 % CI]	p	Holm(2)	$\Delta(G-ctrl)$ [95 % CI]	p	Holm(2)
All- instances det@0.5	441	85.12	88.05	87.11	-2.936 [-3.817 ; -2.122]	<0.0001	<0.001	-1.994 [-2.696 ; -1.328]	<0.0001	<0.001
Individual recall	440	76.74	81.11	80.10	-4.376 [-4.935 ; -3.823]	<0.0001	<0.001	-3.361 [-3.853 ; -2.866]	<0.0001	<0.001
Individual det@0.5	440	85.17	88.08	87.18	-2.903 [-3.819 ; -2.069]	<0.0001	<0.001	-2.002 [-2.712 ; -1.302]	<0.0001	<0.001
Crowd recall	63	69.97	75.71	74.58	-5.736 [-7.864 ; -4.033]	<0.0001	<0.001	-4.612 [-6.531 ; -2.940]	<0.0001	<0.001
Crowd det@0.5	63	82.01	86.24	83.60	-4.233 [-7.937 ; -1.058]	0.0046	0.0092	-1.587 [-4.233 ; +0.529]	0.2250	0.2250
T1 (small- est) recall	315	60.91	66.56	64.99	-5.645 [-6.650 ; -4.627]	<0.0001	<0.001	-4.080 [-5.092 ; -3.091]	<0.0001	<0.001
T1 det@0.5	315	68.26	72.84	71.73	-4.574 [-6.218 ; -2.890]	<0.0001	<0.001	-3.466 [-5.017 ; -1.960]	<0.0001	<0.001
T2 recall	320	80.88	85.46	84.53	-4.584 [-5.316 ; -3.880]	<0.0001	<0.001	-3.656 [-4.370 ; -2.938]	<0.0001	<0.001
T2 det@0.5	320	91.08	93.78	93.15	-2.695 [-3.893 ; -1.609]	<0.0001	<0.001	-2.064 [-3.196 ; -0.928]	0.0004	0.0004
T3 (largest) recall	335	91.68	93.85	93.47	-2.175 [-2.478 ; -1.875]	<0.0001	<0.001	-1.798 [-2.090 ; -1.510]	<0.0001	<0.001
T3 det@0.5	335	98.78	98.96	98.96	-0.180 [-0.697 ; +0.238]	0.4654	0.4654	-0.177 [-0.468 ; +0.054]	0.1586	0.3172
Fragments (comp./img)	500	933.88	615.13	519.26	+318.8 [+305.4 ; +331.9]	<0.0001	<0.001	+414.6 [+397.8 ; +431.1]	<0.0001	<0.001

The trade-off in one sentence: the term **buys** contour quality (Boundary F1 +0.63 vs B, Holm-significant) and thin-object pixel coverage (§6.2), and **pays** in every pedestrian-instance integrity metric — strict recall -4.34, smallest tercile -5.65, crowd groups -5.74, pixel precision -2.70, detection@0.5 -2.94 (all vs paired B, Holm(2) < 0.001 except crowd det@0.5 at 0.0092). The payment is ordered by instance size: T1 -5.65 > T2 -4.58 > T3 -2.18 — **the smallest instances lose the most coverage**, the exact opposite of the term’s intent. Person IoU does rise (+0.344, Holm(2)-significant vs B): the pixel *mass* of the person class grows while the *integrity* of individual pedestrian masks degrades — IoU and instance recall move in opposite directions, which is precisely why the mIoU-centric reading alone would have missed the effect.

6.4 Fragmentation: the mechanism the tables show

New per-class measurement of this paper (500 images × 3 seeds, arms G / B / control, 8-connected components per class per image, paired bootstrap + 19-class Holm; full table in Appendix B):

Reading	Value (G vs B)
Total components per image	933.9 vs 615.1 — ×1.52 (+318.8 [+305.4 ; +331.9], Holm < 0.001)
Classes with more components	16 of 19 vs paired B (14 Holm-significant rises); 19 of 19 vs the control arm. The three that do not rise vs B: terrain -1.8 (Holm < 0.001, the only material decrease), wall -0.02 (p = 0.95) and bus -0.02 (p = 0.70), both null
Largest absolute rises	car +49.4, pole +45.4, vegetation +43.8, road +38.6, traffic sign +38.4, building +32.5, sidewalk +30.8
Pedestrian masks	person 24.3 → 53.2 components/image (×2.2); rider 2.0 → 4.1
Thin classes that <i>gain</i> IoU	also gain components: traffic light +0.9, bicycle +2.2 (Holm-significant)

Revision note (house discipline: the paper reports what the tables show). The program’s initial working hypothesis — inherited from the BRATS intuition that an instance-balancing term suppresses small spurious components in favour of compact thin objects — is **contradicted** by this decomposition: components *increase* almost everywhere, including in the classes whose pixel IoU improves. The mechanism the measurements support is the opposite of compaction: in the exclusive-softmax regime, the equal-weight-per-instance gradient buys **dispersed pixel coverage** for thin and ample classes (per-class IoU ↑, Boundary F1 ↑) at the price of **holes inside ground-truth instances** — a holed mask splits into disjoint components (fragments ×1.52), which mechanically lowers instance-level recall (strict -4.34, T1 -5.65) and pedestrian pixel precision (-2.70). No gradient-level analysis is performed here; §8 lists this as a limitation.

6.5 Versatility ranking: a dominated specialist

The program’s pre-registered versatility criterion: 36 endpoints (19 per-class IoU + 17 business metrics) × 13 arms, same holdout and bootstrap; “maximal damage” = the worst endpoint-level Δ vs control (closer to 0 = no weak point). G’s row and its neighbours (full ranking, artifact results/moe_v3_cs/p317_polyvalence/table_polyvalence.json):

Rank	Arm	Mean percentile	Worst rank	#1	Max damage (pt)	z	Max peak (pt)	ΔmIoU (p / Holm)	Gains sig. (raw/Holm)	Losses sig. (raw/Holm)	Pareto
1	MoE-V3-CS	60.4	11	0	-0.53 (crowd det@0.5)	-0.35	+4.24 (truck IoU)	+0.45 (0.0066 / 0.092)	7/1	2/1	non-dominated
5	B · CE+Dice	51.2	13	2	-4.06 (ped precision)	-0.63	+2.65 (crowd det@0.5)	-0.08 (0.6592 / 1.000)	11/8	6/6	dominated
8	G · CE+Dice+Blob (this paper)	41.9	13	7	-6.76 (ped precision)	-1.05	+3.43 (truck IoU)	+0.09 (0.6592 / 1.000)	4/1	13/12	dominated
11	D · CE+Kervadec EDT	77.3	13	16	-22.20 (ped precision)	-3.44	+5.49 (wall IoU)	+0.52 (0.0050 / 0.075)	17/13	4/3	non-dominated

G’s profile: mean percentile **41.9** (mean rank 8.2), worst rank **13/13**, last on 16 of 36 endpoints — yet **#1 on 7 of 36 endpoints**, more than any arm except D (16): a genuine specialist. The damage is concentrated on the pedestrian family (worst endpoint: pedestrian pixel precision **-6.76 pt**, $z = -1.05$ inter-arm σ ; worst family mean: pedestrian -5.09; best family: contours +0.31). The **price per mIoU point** — maximal damage divided by ΔmIoU — is **-71.7 pt** for G against **-1.2 pt** for the MoE-V3-CS: the mixture buys the same plateau without the hole. G is **Pareto-dominated** (by B, C, C⊙B, Dp⊙B, the control and MoE-V3-CS). The pre-registered T0 verdict (significant ΔmIoU *and* no endpoint > 1 pt below reference *and* no class degraded > 0.5 pt) is reported **as computed: True for the mixture, and G is its antithesis** — a dominated specialist whose speciality (thin objects) is exactly what the mixture absorbs.

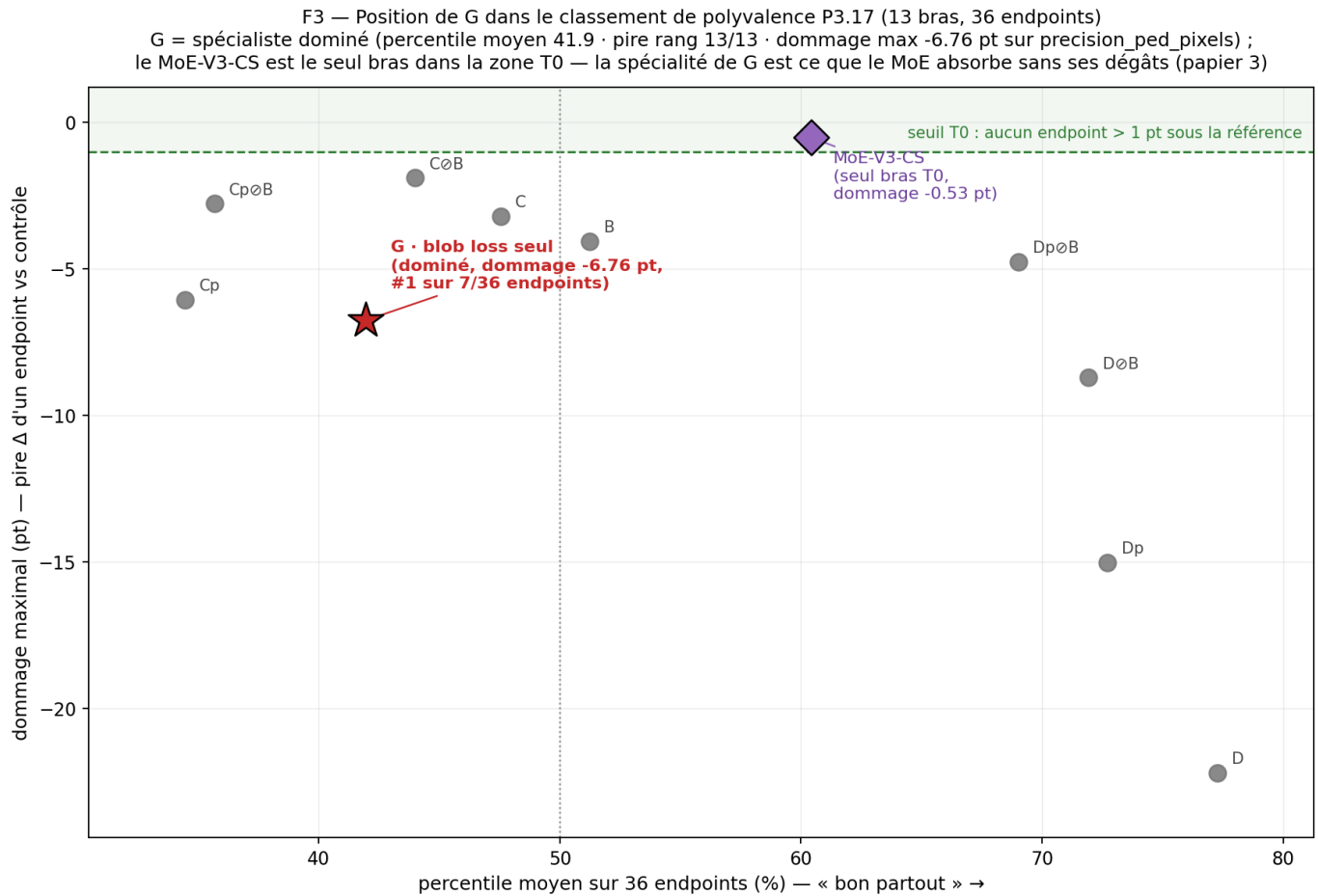


Figure 3: F3 — G in the versatility ranking

Figure 3: position of arm G in the 36-endpoint $\times$ 13-arm ranking, against the mixture MoE-V3-CS and the plateau. Specialist peak (#1 on 7 endpoints) and maximal damage (-6.76 pt) on the same arm.

6.6 Cross-dataset: the same verdict on BRATS

Numbers **cited** from the published BRATS program (second dataset, second probability regime — region-based sigmoid, MedNeXt, nnU-Net recipe, 5-fold CV over $n = 1\,196$; sources verified 2026-09-30, not re-measured here):

BRATS arm	Result	Source
blob alone vs nnU-Net baseline	-0.00376 Dice (5-fold CV; 0/5 folds won; last of the 4 expert arms)	BRATS papers/paper3/versions/v3/NOTE_DECISION_V3 l.46 + tables/v3_vs_consensus.md l.38
blob as expert 3 of the winning MoE-V3	gate V3: +0.00566 Dice vs its own baseline, 1st of 29 arms (5/5 folds)	same note l.52, l.115; [DOI 10.5281/zenodo.22903668]

Two independent programs — different datasets, architectures, metrics and probability regimes — converge on the same verdict: **null-or-negative alone, first-of-the-field as an initialised expert of a mixture**. That convergence is this paper’s generality argument.

7. Discussion

7.1 Why equal weight per instance does not pay alone, in this regime

The measurements support a coherent reading, offered as a hypothesis grounded in the tables (no gradient-level evidence is collected here). Under **exclusive softmax**, every probability mass given to class c inside instance n is taken from the other 18 classes at the same pixel. The per-instance term weights each instance’s *presence* equally, so its gradient keeps pushing class- c mass onto small instances long after the region-wise losses (CE-ISNS, Dice) have settled — mass that lands as **dispersed coverage**: enough scattered correct pixels to raise thin-class IoU (+0.98 traffic light) and contour F1 (+0.63), not enough to keep small masks **connected**. The per-class fragment decomposition (§6.4) shows the cost: holes multiply inside ground-truth instances of 16 classes of 19 vs the paired arm B — of 19 of 19 vs the control (person $\times 2.2$), and a holed mask loses instance recall (−4.34 strict), small-instance recall first (T1 −5.65 > T2 −4.58 > T3 −2.18) and pedestrian pixel precision (−2.70). The class-level mIoU integrates the scattered gains and the scattered losses into a **null** (+0.170, $p = 0.4032$): the term redistributes correctness, it does not create it. On BRATS (sigmoid, independent channels), the same term was also null alone — the competition regime is not the whole story, but the *usage* verdict replicated exactly.

7.2 Where it pays: an initialised expert of a mixture

The companion Cityscapes mixture **MoE-V3-CS** initialises four experts from the independently trained arms B, D, Dp, G and trains a top-2 gate over 3×3 patches for 80 epochs (control: CE+Dice 80 epochs, B-initialised). Measured in the companion study (cited here, artifacts of the same program chain): ΔmIoU **+0.45 pt** [**+0.11 ; +0.82**], **$p = 0.0066$** vs control; small-instance recall **T1 +0.80 (significant)**; traffic light **not degraded** (the boundary/distmap arms lose 2.3–2.8 on it); maximal damage **−0.53 pt**; fragments 488/image — *below* the control’s 519. In other words, the gate routes the thin-object speciality of G to the patches where it wins, and routes away from G where its hole-making behaviour would cost — the speciality is absorbed, the damage is not inherited. The BRATS MoE-V3 did the same with the same expert (1st of 29 arms, 5/5 folds). **The blob loss is a good expert and a bad generalist** — and mixture-of-experts training is the mechanism that converts one into the other.

7.3 Practical takeaways

- **Do not ship the blob term alone** in exclusive-softmax semantic segmentation at fixed budget: expect a null mIoU, better thin-object pixel IoU, and degraded pedestrian instance integrity. If pedestrian *detection* feeds a downstream planner, the −4.3 strict-recall points are disqualifying on their own.
- **Do consider it as an expert candidate** for a patch-wise mixture: it is the only arm of the 13 that improves traffic light vs the control, and its speciality is exactly what the measured mixture absorbs.
- **If you keep it, pair it with a CC filter**. The fragmentation diagnosis (+318.8 components/image) says a post-hoc connected-component consensus veto — the mIoU-neutral $C\oslash B$ of [Cassez 2026b], −18.7 % fragments — targets precisely this arm’s failure mode. (Reported as a design implication; the $G\oslash$ -style veto on this arm is not measured in this paper.)
- **The engineering cost is solved**: precomputed instance packs make the term +1–2 %/epoch (against +870 s/epoch naive), parity-tested, with the flip-alignment pitfall documented (§4).

7.4 Relation to Kofler et al.

Nothing here contradicts the published medical result: under the region-based sigmoid recipe the term improved the small-structure metrics it targets [Kofler 2023]. What this paper adds is the **regime boundary**: transposed *exactly* (same algebra, same $\beta = 0.5$, same two-level mean) to exclusive-softmax full-resolution Cityscapes, the term’s promise inverts into the trade-off above. Loss-term transfers across probability regimes should be measured, not assumed — on both of our datasets the *alone* verdict was null while the *expert* verdict was first-of-the-field.

8. Limitations

1. **The primary endpoint is null**. This is a negative-result paper by design; the contribution is the diagnostic, the port and the usage — not a performance claim.
2. **β not swept**. $\beta = 0.5$ fixed at Kofler’s default ($\alpha:\beta = 2:1$); no λ sweep, same fixed-budget stance as the program’s companion papers. The trade-off could move with β ; that surface is not measured.
3. **Holdout first:500**. A pre-specified subset of val, identical for all 13 arms and fixed before training — but not the official leaderboard, which was not submitted to.
4. **Holm on the 15-pair exploratory family leaves no arm significant on mIoU** (best raw $p = 0.0048 \rightarrow$ Holm 0.058). Program-wide conclusions on mIoU rest on amplitudes and CIs, stated as such.

5. **Budget mismatch with the 13-arm context.** G trains 160 epochs; the context control 80. The paired comparison of G is B (160 epochs); the control pair is always labelled *context*.
6. **Mechanism is a measurement-grounded hypothesis.** The fragmentation decomposition is new and blocking-checked, but no gradient-level or per-layer analysis is performed; §7.1 remains an interpretation.
7. **n = 3 seeds.** Sub-0.5-pt effects are underpowered at the seed level; the primary inference is the 500-image paired bootstrap, which probes evaluation-set sampling, not seed variance.
8. **cuDNN/BF16 non-determinism declared.** Two independent forwards of the same checkpoints differ by ≤ 0.01 pt (verdicts identical); bit-exact reruns are not claimed anywhere in the program.
9. **One architecture, two datasets.** ConvNeXt-V2-Base + UPerNet only; Cityscapes + BRATS only. Other heads (Mask2Former-style mask attention re-weights boundaries internally) may react differently.
10. **Instance packs not shipped** (heavy); regenerated bit-identical by the released script, parity verified over all 2 975 packs on 2026-09-26.

9. Conclusion

Given every ground-truth instance the same weight, whatever its pixel count, and full-resolution exclusive-softmax Cityscapes answers: **not worth it alone**. The pre-registered primary endpoint is null ($\Delta\text{mIoU} +0.170$ pt [-0.233 ; $+0.551$], $p = 0.4032$, G vs paired B, 160 epochs, 3 seeds). The term buys exactly what it promises — thin-object pixel coverage (traffic light $+0.98$, pole $+0.58$, bicycle $+0.57$, Holm-significant; Boundary F1 $+0.63$) — and pays for it in pedestrian instance integrity (strict recall -4.34 , smallest tercile -5.65 , crowd -5.74) and in mask topology (components $\times 1.52$, 16 classes of 19 up vs paired B and 19 of 19 vs the control, person masks $\times 2.2$), which leaves it a **Pareto-dominated specialist** on the program’s 36-endpoint versatility criterion. The same term, as an initialised expert of a patch-wise mixture, is **first of 29 arms on BRATS** and part of the **only non-dominated, significantly-positive arm on Cityscapes** (MoE-V3-CS, $\Delta\text{mIoU} +0.45$, $p = 0.0066$, maximal damage -0.53). Equal weight per instance does not pay alone — it pays as an expert.

Code, configs, per-arm artifacts pointers, tables, figures and regeneration scripts: github.com/guillaume-cassez/cityscape-blob-loss-kofler. Program companions: Cityscapes distmap [DOI 10.5281/zenodo.21006236], Cityscapes boundary ablation [DOI 10.5281/zenodo.21006393], BRATS MoE-V3 [DOI 10.5281/zenodo.22903668], BRATS boundary [DOI 10.5281/zenodo.22906447], BRATS consensus [DOI 10.5281/zenodo.22904810], BRATS distmap [DOI 10.5281/zenodo.20110976].

Appendix A — Runtime and reproducibility

Stage	Hardware	Time	Output
Instance-pack precomputation (one-shot, train split)	8 P-cores, cc3d	22.4 ms/img	.blob.npz per image (blob_glob/counts/csr)
Training 160 ep, 1 seed (arm G)	1 $\times$ RTX PRO 6000 96 GB Blackwell	29.2 h (656 s/ep, 31.5 GB)	checkpoints/pilot_fullres_G_blob_seed{s}/epoch_
Arm G total (3 seeds)	same	≈ 87 h GPU	final train loss 0.3743 / 0.3735 / 0.3765
Primary bootstrap (harness)	CPU	minutes	results/moe_v3_cs/harness/table_Gseul_P310.json
Business metrics + fragments (2 arms $\times$ 3 seeds)	1 GPU + 6 workers	≈ 2.6 h (12-arm job)	metriques_G_seed*.npz, frag_*_seed*.npz
Tables + figures of this paper	CPU	minutes	papers/paper4/{tables,figures}/

Reproducibility seeds. Training seeds 42, 123, 456 are set globally (PyTorch, NumPy, Python random, CUDA). The bootstrap seed is fixed (20260618, B = 10 000) and shared by every table of the program, so all CIs and p-values are exactly re-derivable from the released per-image artifacts. cuDNN benchmark is left **on** (deterministic: false) for training speed; bit-exact GPU reruns are therefore not claimed — the declared reproducibility statement is: same artifacts $\rightarrow$ same statistics bit-for-bit (verified), and independent forwards of the same checkpoints $\rightarrow$ agreement within 0.01 pt (measured, 14/14 blocking sanity checks).

Appendix B — Per-class fragmentation, G vs B (components per image)

8-connected components per class per image, 500 images $\times$ 3 seeds, paired bootstrap B = 10 000 (seed 20260618), Holm over the 19-class family. Bold = Holm < 0.05 . New measurement of this paper (the program had fragments only at arm level before it).

Class	G	B	ctrl	$\Delta(G-B)$	95 % CI	p	Holm(19)
car	117.3	67.9	57.4	+49.4	[+46.5 ; +52.2]	<0.0001	<0.001
pole	176.4	131.1	115.9	+45.4	[+42.9 ; +47.8]	<0.0001	<0.001
vegetation	107.7	63.8	58.8	+43.8	[+41.6 ; +46.1]	<0.0001	<0.001
road	82.4	43.8	47.1	+38.6	[+36.3 ; +40.9]	<0.0001	<0.001
traffic sign	92.3	53.9	43.2	+38.4	[+36.1 ; +40.7]	<0.0001	<0.001
building	116.9	84.4	75.4	+32.5	[+29.6 ; +35.4]	<0.0001	<0.001
sidewalk	106.2	75.4	64.6	+30.8	[+28.7 ; +33.0]	<0.0001	<0.001
person	53.2	24.3	16.4	+28.9	[+27.3 ; +30.5]	<0.0001	<0.001
sky	27.7	23.2	11.9	+4.6	[+3.2 ; +5.9]	<0.0001	<0.001
fence	10.9	8.2	4.9	+2.7	[+2.1 ; +3.4]	<0.0001	<0.001
bicycle	11.8	9.6	6.2	+2.2	[+1.7 ; +2.7]	<0.0001	<0.001
rider	4.1	2.0	1.9	+2.2	[+2.0 ; +2.4]	<0.0001	<0.001
traffic light	6.9	6.0	4.6	+0.9	[+0.6 ; +1.1]	<0.0001	<0.001
truck	0.6	0.3	0.3	+0.2	[+0.1 ; +0.4]	<0.0001	<0.001
motorcycle	0.7	0.6	0.6	+0.0	[-0.0 ; +0.1]	0.4940	1.000
train	0.3	0.2	0.2	+0.0	[-0.0 ; +0.0]	0.3292	1.000
bus	0.5	0.5	0.4	-0.0	[-0.1 ; +0.1]	0.7028	1.000
wall	7.2	7.2	4.0	-0.0	[-0.5 ; +0.5]	0.9496	1.000
terrain	10.9	12.8	5.5	-1.8	[-2.6 ; -1.1]	<0.0001	<0.001

Sanity: the per-class sums equal the arm-level component counts **bit-for-bit** (G 933.9, B 615.1, control 519.3 per image). Source: results/moe_v3_cs/paper4_blob/table_frag_classes_G_vs_B.json.

References

- Cordts *et al.* (2016). *The Cityscapes dataset for semantic urban scene understanding*. CVPR.
- Cassez (2026a). *Distance-Map Auxiliary Regression for Full-Resolution Cityscapes Segmentation*. Zenodo. DOI 10.5281/zenodo.21006236.
- Cassez (2026b). *Boundary Loss Ablation for Full-Resolution Cityscapes Segmentation: When Dice Helps and When It Doesn't*. Zenodo. DOI 10.5281/zenodo.21006393.
- Cassez & Larnier (2026c). *The Gate Does Not Choose: An Expert-Initialised Mixture-of-Experts Outperforms 24 Arms on BRATS 2023*. Zenodo. DOI 10.5281/zenodo.22903668.
- Cassez & Larnier (2026d). *Precision Pays: A Fixed-Weight Boundary Loss Anchors the Precision End (BRATS)*. Zenodo. DOI 10.5281/zenodo.22906447.
- Cassez & Larnier (2026e). *Two Models That Agree Beat the Best of Them Alone: Parameter-Free Connected-Component Consensus (BRATS)*. Zenodo. DOI 10.5281/zenodo.22904810.
- Cassez (2025). *Distance Map Auxiliary Loss for Brain Tumor Segmentation: A Fragment-Centric Study*. Zenodo. DOI 10.5281/zenodo.20110976.
- Isensee *et al.* (2021). *nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation*. Nature Methods 18.
- Kervadec *et al.* (2019). *Boundary loss for highly unbalanced segmentation*. MIDL. arXiv:1812.07032.
- Kofler *et al.* (2023). *Blob loss for biomedical image segmentation*. IPMI. arXiv:2205.08209.
- Lin *et al.* (2017). *Focal loss for dense object detection*. ICCV.
- Milletari *et al.* (2016). *V-Net: fully convolutional neural networks for volumetric medical image segmentation*. 3DV.
- Perazzi *et al.* (2016). *A benchmark dataset and evaluation methodology for video object segmentation*. CVPR.
- Silversmith *et al.* (2019). *cc3d: connected-components labeling for large 3D arrays (software)*. github.com/seung-lab/connected-components-3d.
- Woo *et al.* (2023). *ConvNeXt V2: co-designing and scaling ConvNets with masked autoencoders*. CVPR. arXiv:2301.00808.
- Xiao *et al.* (2018). *Unified perceptual parsing for scene understanding*. ECCV. arXiv:1807.10221.

Manuscript — 2026-09-30. Source code, configs, tables, figures and regeneration scripts: github.com/guillaume-cassez/cityscape-blob-loss-kofler. Author: Guillaume Cassez, independent researcher (ORCID 0009-0007-0987-3931), guillaume-cassez.fr — currently looking for ML / computer vision engineering opportunities.